

Information sheet

This information sheet is intended for participants and/or their guardians involved in the research study conducted by Electronics and Telecommunication Engineering students from the University of Moratuwa.

Research title : Beyond Optical PPG: Adaptive mmWave Radar Arterial Pulse Sensing

List of investigators

Principal Investigator: Dr. Sampath Perera (Senior Lecturer, Dept. of Electronic and Telecommunication Engineering, Faculty of Engineering, University of Moratuwa)

Co-Investigators: Dr. Pranjeevan Kulasingham (Senior Lecturer, Dept. of Electronic and Telecommunication Engineering, Faculty of Engineering, University of Moratuwa)

Thamilezai Ananthakumar (Undergraduate, University of Moratuwa)

Kopithan Manoranjithan (Undergraduate, University of Moratuwa)

Mathujan Shanmugavadivel (Undergraduate, University of Moratuwa)

Sivamynthan Nadesamoorthy (Undergraduate, University of Moratuwa)

Introduction

This project aims to develop an adaptive non-invasive heart rate and blood pressure monitoring system using mmWave radar to improve cardiovascular monitoring. The technology is designed to reduce the limitations of conventional optical (PPG) sensors, including ambient light interference, sweat, skin pigmentation, and reduced accuracy across different subcutaneous fat thicknesses.

Objectives

Replacing the limitations of current optical (PPG) based sensing — ambient light, sweat, subcutaneous fat-layer effects, and skin pigmentation — with adaptive mmWave radar-based sensing for heart rate and blood pressure monitoring.

Methodology of the research

Participants:

Approximately 30 healthy adult volunteers ranging in age from 22 to 26 years will be recruited (university of moratuwa students). However, if sufficient variation in skin tone and BMI/subcutaneous fat thickness cannot be achieved within the UoM student population, participants may also be recruited from outside the University. Participation will be entirely voluntary, and written informed consent will be obtained prior to data collection.

Brief Description:

The study involves the simultaneous collection of physiological data using a custom wristband prototype together with two reference devices: a Polar H10 chest strap for heart rate and FDA Approved Omron wrist-cuff monitor for blood pressure. The wristband integrates a 60 GHz mmWave radar, an optical photoplethysmography (PPG) sensor, and an inertial measurement unit (IMU).

Methods of Data Analysis:

Signals will be preprocessed (noise removal, filtering), features extracted (heart rate, blood pressure, and pulse-waveform characteristics), and analysed using Python/MATLAB algorithms. Results will be validated by comparing the wearable sensor outputs against the reference devices — the Polar H10 for heart rate and the Omron BP4350 for blood pressure — using accuracy and reliability metrics.

Potential benefits to the participant and for the wider community

(a) To the participant

Participants may not receive direct medical benefits but may gain awareness of their physiological measurements recorded during the study.

(b) To humanity generally

The study contributes to developing robust, non-invasive cardiovascular monitoring systems that reduce limitations of conventional optical sensors, such as ambient light interference, skin pigmentation, and tissue variation effects.

Data collection procedure

Basic demographic information (age, gender, height, weight, skin tone/pigmentation, and subcutaneous fat thickness) will be collected for analysis purposes. The Polar H10 chest strap, the Omron wrist-cuff monitor, and the multimodal wristband prototype will be fitted to the participant for non-invasive physiological signal

acquisition. The wristband will be placed on the participant's wrist over the radial artery. Participants will perform activities such as sitting, walking, and running while physiological signals are recorded. Data will be collected simultaneously from the wearable prototype and the reference devices.

Risks, hazards, side effects and discomforts

Participants will come into contact with a wearable prototype containing an ams-osram optical front-end and an Infineon 60 GHz radar module, together with a Polar H10 chest strap and an Omron wrist-cuff monitor. The radar transmits a low-power electromagnetic wave toward the tissue to measure mechanical deformation from arterial pulsation. This constitutes safe, low-power non-ionizing radiation that is standard for commercial near-field sensing. No electrical current is applied to the participant. The study involves only non-invasive measurements and is considered minimal risk.

Participant selection

Inclusion Criteria: Adults aged 22-26 years who are healthy and capable of providing informed consent.

Exclusion Criteria: Individuals with diagnosed cardiovascular diseases or conditions affecting peripheral circulation

Participants will primarily be recruited from the University of Moratuwa community through convenience sampling. However, if sufficient variation in skin tone and BMI/subcutaneous fat thickness cannot be achieved within the UoM student population, additional participants may be recruited from outside the University through convenience sampling. Interested volunteers will be screened for eligibility and provided with study information before participation.

Voluntary participation and rights

Participation in this study is entirely voluntary. Participants may choose to withdraw from the study at any time without any consequences or impact on medical care.

How the data will be used

The data collected in this study may be used for academic and scientific purposes, including research publications in peer-reviewed journals, conference proceedings, and academic presentations. The findings may also contribute to recommendations for future research and the development of wearable health-monitoring technologies. No personally identifiable information will be included in any publication or presentation.

Confidentiality and data security

All collected data will be kept strictly confidential. Personal identifiers will be removed, and the participant's identity will be anonymized. Data will be securely stored on a password-protected system and in the University repository.

The results may be published in academic journals, presented at conferences, or included in student theses. All publications will use anonymized data only.

Compensation

No financial or material compensation will be provided for participation.

Clarifications and complaints

If you have any questions or need more information, please contact:

- Dr. Joshua Pranjeevan Kulasingham – pranjeevank@uom.lk
- Thamilezai Ananthakumar - ananthakumart.22@uom.lk
- Kopithan Manoranjithan - kopithanm.22@uom.lk
- Mathujan Shanmugavadivel - mathujans.22@uom.lk
- Sivamynthan Nadesamoorthy - sivamynthann.22@uom.lk